Market making: turning the signal into money you can keep

Learning notes for `src/market_maker.cpp`. The taker backtest ended on a cliffhanger: order-book imbalance predicts direction, but the move is smaller than the spread, so you cannot profit by *crossing* it. The natural question is the one a market-making desk lives on: what if you *provide* the spread instead of paying it? This is that experiment, and it closes the loop.

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1. The two ways to trade a signal

- **Taker** (backtest.cpp): cross the spread to get in now. You pay ~1 tick round-trip and need the move to exceed it. OBI's move does not, so a taker loses.
- **Maker** (this file): post a bid and an ask and wait to get hit. Now the spread is your *income*, not your cost. But you inherit a new enemy: **adverse selection** - the informed traders pick you off, buying from you right before a rise and selling to you right before a fall. A maker's whole job is to earn the spread faster than adverse selection bleeds it back.

The signal changes role too: a taker used OBI to *pick a direction*; a maker uses it to *decide which quote to defend*.

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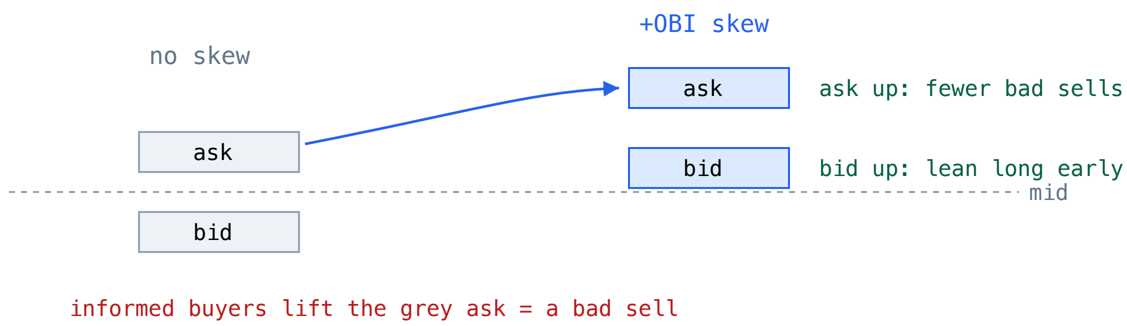
2. The reservation price

We quote around a reservation price, not the raw mid (the Avellaneda-Stoikov idea):

```
r = mid  -  gamma * inventory  +  theta * OBI
             \-----------/          \-------/
           inventory skew          signal skew
  bid = r - delta        ask = r + delta      (delta = base half-spread)
```

- **Inventory skew** (`- gamma * inventory`): if we are long, `r` drops, pulling *both* quotes down - our ask gets hit more (we sell down the position), our bid less. This mean-reverts inventory toward zero. It is pure **risk management**, and a hard position limit `+/- Qmax` backstops it.
- **Signal skew** (`+ theta * OBI`): when imbalance says price is about to rise, `r` rises, lifting our ask out of the way of informed buyers (fewer bad sells) and our bid up so we lean gently long *before* the move. When it says fall, the mirror. This is the signal defending the inventory.

OBI > 0: skew the whole quote UP to dodge informed buyers



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3. Making adverse selection real

A backtest that assumes fills are random would show a maker printing free spread forever - useless. The fills here are generated from the loblab sim so the danger is real. Fill intensity on each side **decays with quote distance** (further out, less likely to trade) and **tilts with the hidden informed pressure** (`FV − mid`):

```
double up = clamp(presc * pressure, -0.95, 0.95);     // FV − mid: the informed direction
double bid_rate = base * exp(-k*bidDist) * (1 − beta*up);   // fewer bids hit when price rising
double ask_rate = base * exp(-k*askDist) * (1 + beta*up);   // more asks hit when price rising
if (inv >=  Qmax) bid_rate = 0;   // position limit
if (inv <= -Qmax) ask_rate = 0;
```

The crucial asymmetry: **the maker never sees** `FV` (the truth driving informed flow). It only sees `OBI` , a noisy proxy. So the question is whether skewing on that proxy actually helps.

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4. The result (A/B over the same path and fill luck)

```
A: inventory only    PnL +384544 t    adverse +65986    avg|inv| 8.3    spread-retention 85.4%
B: inventory + OBI   PnL +396911 t    adverse +49982    avg|inv| 6.9    spread-retention 88.8%
```

Same market, same random fills - the only difference is whether the maker skews on OBI. Doing so:

- lifts **PnL +3.2%** (+12,367 ticks),
- cuts **adverse selection ~24%** (65,986 -> 49,982),

- *lowers* average inventory (8.3 -> 6.9), so it earns **more while holding less risk**,
- raises **spread retention** (share of captured spread kept after adverse selection) 85.4% -> 88.8%.

Every axis improves at once. That is the signature of a real edge, not a variance artifact.

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5. The punchline that ties the repo together

> The *same* order-book-imbalance signal that **loses money crossing the spread** (taker backtest) **makes money providing it** (this market maker).

That is not a contradiction - it is the whole lesson of microstructure alpha. A signal's value is inseparable from *how you trade it*. Crossing the spread, OBI's edge is smaller than the toll. Quoting the spread, the same edge becomes a way to keep more of the spread you already earn, by stepping out of the way of informed flow. Being able to show both halves - and connect them - is worth more than either number alone.

Caveats (honest, as before): synthetic latent-fair-value flow, a simplified fill model (no explicit queue position), and PnL in ticks without fees or a live spread. The transferable result is the *framework and the direction of every effect*; the magnitudes must be re-earned on recorded data with a queue model. Next steps live in the README roadmap.